FORM CaseCraft

Complete site documentation & image upload size recommendation guide

Generated June 30, 2026 · FORM Case Studio · Private documentation

Table of Contents

1. 1. Platform Overview
2. 2. Image Upload Size Guide (All Types)
3. 3. Image Preview Examples
4. 4. Storefront Pages & Routes
5. 5. Admin Dashboard
6. 6. Order & Payment Flow
7. 7. API & Technical Reference
8. 8. Design System

1. Platform Overview

FORM CaseCraft is a full-stack custom phone case e-commerce platform. Customers can design their own cases or buy pre-made templates from the Shop. Orders are paid via UPI with manual screenshot verification.

Core Capabilities

Area	Features
CUSTOMER	Custom designer (Fabric.js), template shop, user accounts, checkout, UPI payment, order tracking, wishlist
ADMIN	Orders, catalog (brands/models/case types), case overlay templates, shop template products, promo codes, popup offers, settings
Backend	Node.js + Express + MongoDB, JWT auth, Multer file uploads (max 12 MB general), static serving at `/api/uploads/`

Tech Stack

- **Frontend:** React 19, React Router 7, Tailwind CSS, Shadcn UI, Fabric.js, Framer Motion
- **Backend:** Node.js, Express, MongoDB, Multer, JWT, bcryptjs
- **Payments:** Manual UPI (QR code + screenshot + UTR verification)

2. Image Upload Size Guide

All uploads accept common web image formats unless noted. Server enforces hard file-size limits — stay under the recommended sizes for faster uploads and better performance.

Global allowed formats (most uploads): JPG, JPEG, PNG, WEBP, GIF

Complete Upload Specifications

Upload Type	Who Uses It	Recommended Dimensions	Aspect Ratio	Formats	Max File Size	Notes
Customer design artwork `/upload/design`	Customer (Customizer)	2000 × 4000 px or higher *(portrait, high-res)*	~9:16 phone portrait	JPG, PNG, WEBP	12 MB	Drag, zoom, rotate on Fabric.js canvas. Clipped to camera mask zones. Use high-res source for sharp prints.
Canvas preview PNG `/upload/preview`	System (auto on checkout)	Matches template canvas *(e.g. 360×720 – 1080×2160)*	Phone case frame	PNG	12 MB	Generated automatically when customer continues to checkout. Stored with order.
Payment screenshot `/upload/payment`	Customer (Payment page)	Any readable screenshot *(phone screen capture)*	Any	JPG, PNG, WEBP, GIF	12 MB	UPI payment proof. Must show amount and transaction reference clearly.
Case overlay template `/admin/templates`	Admin (Case Overlays)	Native PNG dimensions *(typically 800–1200 px tall)*	Phone outline	**PNG only**	12 MB	Plain phone frame PNG. Canvas size read from image. Draw camera/flash mask zones in Zone Editor.

Shop template thumbnail `/admin/template-products`	Admin (Templates shop)	`800 × 640 px`	5:4	JPG, PNG, WEBP	`12 MB` *rec. 2 MB*	Center the phone case. Dark (#111) or transparent background. Matches shop card frame.
Shop template gallery `/admin/template-products`	Admin (Templates shop)	`800 × 640 px` each	5:4	JPG, PNG, WEBP	`12 MB` each *rec. 2 MB*	Up to 10 gallery images per product. Same spec as thumbnail.
Brand logo `/admin/catalog`	Admin (Catalog)	`128 × 128 px`	1:1 square	JPG, PNG, WEBP, GIF	`2 MB`	Shown in brand picker during customization.
Phone model image `/admin/catalog`	Admin (Catalog)	`400 × 400 px`	1:1 square	JPG, PNG, WEBP, GIF	`4 MB`	Model selection cards in customizer.
Case type image `/admin/catalog`	Admin (Catalog)	`400 × 400 px`	1:1 square	JPG, PNG, WEBP, GIF	`4 MB`	Silicone, matte, transparent, glass case type cards.
Homepage popup offer `/admin/popup-offers`	Admin (Marketing)	`800 × 1000 px`	4:5 portrait	JPG, PNG, WEBP, GIF	`12 MB`	Promotional modal on homepage. Schedule with start/end dates.

Quick Reference — Recommended Sizes

Purpose	Size	Ratio
Shop / template product cards	800 × 640 px	5:4
Popup promotional banner	800 × 1000 px	4:5
Brand logo	128 × 128 px	1:1
Model & case type images	400 × 400 px	1:1
Customer print artwork	2000+ px tall (portrait)	~9:16
Case overlay frame PNG	Native phone outline PNG	Varies by model

Best Practices

- **Shop thumbnails:** Center the phone case on a dark (#111111) or transparent background. Avoid busy backgrounds — cards crop to 5:4.
- **Customer artwork:** Upload the highest quality source you have. The canvas scales down for preview but the original is used for printing.
- **Case overlay PNGs:** Use a clean phone outline with transparent areas for camera cutouts. PNG only — no JPEG.
- **Compress before upload:** Use TinyPNG, Squoosh, or export at 80–90% JPEG quality to stay under recommended sizes without visible quality loss.
- **Payment screenshots:** Ensure UTR/reference number and amount are legible. Full-screen UPI success screen works best.

3. Image Preview Examples

Sample images from the project showing how each upload type appears in the system.

Static Site Assets (Public)



Site Logo
case_box.png — Header & favicon



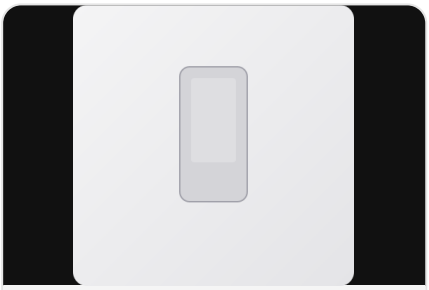
Frame Thumbnail
frame-tum.jpg — Showcase frame



404 Hero
404-hero.png — Not found page

Catalog Images (Admin Uploads)

Brand Logo
128×128 px recommended · max 2 MB



Phone Model
400×400 px recommended · max 4 MB



Case Type
400×400 px recommended · max 4 MB

Case Overlay Templates (PNG Frames)



Phone Frame PNG
PNG only · mask zones for camera



Case Overlay
Canvas size from PNG dimensions



Transparent Frame
Used in Fabric.js customizer

Shop Template Products (5:4 Thumbnails)



Shop Thumbnail
800×640 px · 5:4 ratio

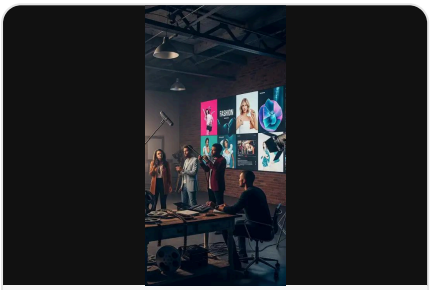


Template Card
Dark background recommended



Gallery Image
Same 800×640 spec

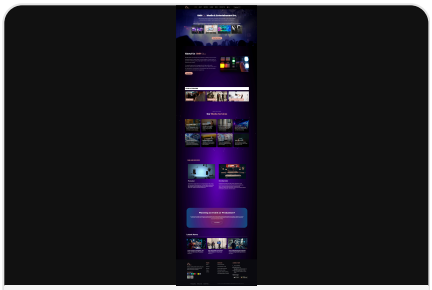
Customer Designs & Payments



Customer Artwork
High-res portrait · max 12 MB



Design Upload
JPG · PNG · WEBP



Payment Screenshot
UPI proof · max 12 MB

4. Storefront Pages & Routes

Route	Page	Description	Auth
/	Home	Video hero, brand marquee, product showcase, collections, best sellers, FAQ, customize CTA	Public
/shop	Shop	Browse pre-designed template cases. Filter by category, brand, case type, collection, sort	Public
/template/:slug	Template Detail	Product page with gallery, phone model picker, add to cart / checkout	Public
/customize	Customize	4-step wizard: Brand → Model → Case type → Design (Fabric.js canvas)	Public
/checkout	Checkout	Shipping form (Indian states, pincode), promo code, order creation	Login required
/payment/:orderId	Payment	UPI QR, copy UPI ID, screenshot upload, UTR entry, WhatsApp link	Login required
/track	Order Status	Search by order ID	Public
/track/:orderId	Order Status	Delivery timeline, order details, design preview	Public
/login	Login	Customer email/password sign-in	Public
/register	Register	Create customer account	Public
/profile	Profile	Account details, order history	Login required
*	404 Not Found	Custom 404 page with hero image	Public

Customizer Flow (4 Steps)

1. **Choose brand** — Apple, Samsung, OnePlus, Vivo, Oppo, Xiaomi, Google Pixel
2. **Choose model** — Exact phone model (e.g. iPhone 15 Pro Max)
3. **Choose case type** — Silicone (₹299), Transparent (₹349), Matte (₹399), Glass (₹499)
4. **Design** — Upload artwork, drag/zoom/rotate on live canvas with phone frame overlay

Shop Template Categories

Anime, Gaming, Luxury, Minimal, Nature, Abstract, Sports, Travel, Cars, Quotes, Photography, Premium

Collections

Essentials, Limited Edition, Artist Series, Seasonal, Collab

5. Admin Dashboard

Access at `/admin/login`. JWT authentication with HTTP-only cookies. Seeded from `ADMIN_EMAIL` and `ADMIN_PASSWORD` in backend `.env`.

Route	Section	What You Manage
`/admin`	Dashboard	Analytics: total orders, pending review, approved, shipped, revenue. Recent orders table.
`/admin/orders`	Orders	Filter by status, approve/reject payments, update delivery stages, view design & payment screenshots
`/admin/template-products`	Templates (Shop)	Pre-designed cases for sale: title, price, category, thumbnail (800×640), gallery, stock, SKU
`/admin/catalog`	Brands & Products	Phone brands (128×128 logo), models (400×400), case types (400×400 + price)
`/admin/catalog/templates`	Case Overlays	Phone frame PNG templates with camera mask zones per brand/model/case type
`/admin/promo-codes`	Promo Codes	Discount codes: percentage or flat amount, usage limits, expiry
`/admin/popup-offers`	Popup Offers	Homepage promotional popups (800×1000 px), schedule, enable/disable
`/admin/settings`	Settings	UPI ID, merchant name, WhatsApp number, currency

6. Order & Payment Flow

> `Home → Shop or Customize → Checkout → Payment (UPI) → Track Order`

Custom Order Path

1. Customer selects brand, model, case type in Customizer
2. Uploads artwork and positions on Fabric.js canvas
3. Design + preview PNG uploaded to backend; draft saved in sessionStorage
4. Customer fills shipping at Checkout → order created (ID: `FRM-YYMMDD-XXXXXX`)
5. Customer pays via UPI, uploads screenshot, enters UTR
6. Admin verifies payment in Orders panel
7. Customer tracks delivery on Order Status page

Template Shop Path

1. Customer browses Shop, picks pre-designed template
2. Selects phone model on Template Detail page
3. Checkout → Payment → Track (same as custom)

Payment Statuses

Status	Meaning
`pending`	Order created, no payment proof yet
`submitted`	Customer uploaded screenshot and UTR
`approved`	Admin verified payment
`rejected`	Payment rejected by admin

Delivery Statuses

Status	Meaning
`pending_payment_verification`	Awaiting payment review
`payment_approved`	Payment confirmed
`printing`	Case is being printed
`packed`	Order packed
`shipped`	Out for delivery
`delivered`	Delivered to customer

7. API & Technical Reference

Base URL: `{BACKEND_URL}/api` (default: `http://localhost:8000/api`)

Upload Endpoints

Method	Endpoint	Auth	Purpose
POST	`/upload/design`	—	Customer artwork
POST	`/upload/preview`	—	Canvas preview PNG
POST	`/upload/payment`	User	Payment screenshot

Static files served at:
`/api/uploads/{designs|payments|templates|brands|models|products|template-products|popups}/<filename>`

Environment Variables

Backend (.env)

- `MONGO_URL` — MongoDB connection
- `DB_NAME` — Database name
- `JWT_SECRET` — Token signing
- `ADMIN_EMAIL` / `ADMIN_PASSWORD`
- `PORT` — default 8000
- `PUBLIC_BASE_URL` — upload URL prefix

Frontend (.env)

- `REACT_APP_BACKEND_URL` — API base URL

Getting Started

```
# Backend
cd backend && npm install && npm start

# Frontend
cd frontend && npm install && npm start

# Admin login at http://localhost:3000/admin/login
```

8. Design System

Token	Value	Usage
Primary	#FF4F00	CTAs, accents, highlights
Background	#09090B	Page background
Text primary	#FFFFFF	Headings, body on dark
Text secondary	#A1A1AA	Supporting text
Success	#10B981	Confirmations
Error	#EF4444	Errors, delete actions

Typography

- **Headings:** Clash Display
- **Body:** Satoshi
- **Labels / mono:** JetBrains Mono

UI Patterns

- **Crystal glass:** Frosted glass cards with subtle borders
- **Border radius:** rounded-2xl for cards, rounded-lg for buttons
- **Icons:** Lucide / Phosphor for premium hardware feel
- **Theme:** Dark luxury (Archetype 5) — no purple/teal accents

FORM CaseCraft · Complete Documentation · © FORM Case Studio · All rights reserved